OpenLecture:A Distributed Platform for Global Academic Lectures

1 Description of the Use Case

Many universities have highly reputed professors whose courses attract interest far beyond their own institutions. However, access to these lectures is typically limited to students enrolled in that university. While lecture recordings may occasionally be shared online, current classroom systems rarely support **live distributed classrooms where students from multiple institutions can attend a lecture simultaneously and interact with instructors in real time**.

The proposed system, **GlobalClass**, is a distributed classroom platform designed to enable **scalable lecture streaming to a large number of students across universities**. Professors from participating institutions can host lectures that students from other universities can attend through a unified system. The platform focuses on delivering lectures to large audiences while maintaining reliable performance and enabling interactive participation.

Stakeholders

- Students seeking access to lectures from top professors across institutions
- Instructors delivering courses to a larger academic audience
- Universities sharing academic resources and course offerings
- Administrators managing courses and monitoring system usage

Problem Statement

Existing classroom systems are primarily designed for single institutions or small class sizes. They struggle to support distributed learning environments where thousands of students may attend a lecture simultaneously. Key challenges include handling large numbers of concurrent viewers, enabling real-time interaction between instructors and students, and managing access across multiple institutions.

When a lecture by a well-known professor becomes popular, the system must reliably deliver the lecture stream to many users while maintaining low latency and interactive capabilities.

2 Key Functionalities

Core Feature

- Scalable live lecture streaming to large concurrent audiences

Supporting Features

- Global course catalog across institutions
- Cross-institution lecture enrollment
- Student question submission and voting
- Lecture recordings and playback
- Notifications for lecture updates
- Lecture participation analytics

User Interaction

Students browse courses, enroll in lectures, join live streams, and submit questions. Instructors create and manage lectures and respond to student queries. Administrators monitor system usage and lecture analytics through dashboards.

Technical Highlights

- **Frontend: React** – Responsive UI for live lecture dashboards and interaction.
- **Backend: Node.js** – Event-driven architecture suitable for handling large concurrent users.
- **Database: PostgreSQL** – Reliable relational storage for users, courses, and enrollments.
- **Streaming Technology: WebRTC** – Enables low-latency real-time video streaming.

Architectural Tactics

- Scalability through microservices architecture
- Load balancing and caching for high-traffic lectures
- Fault isolation to improve system availability
- Role-based authentication and authorization
- Modular design to support extensibility

Design Patterns

- Observer Pattern – notifications and real-time updates
- Strategy Pattern – ranking and prioritizing student questions
- Factory Pattern – creation of lecture session objects
- Repository Pattern – abstraction of database access

3 Expected Time to Build a Prototype

A team of five members will design the complete system architecture while implementing **2–3 key end-to-end functionalities** such as lecture creation, lecture streaming access, and question submission.

Research and Planning (3–4 days) Study existing platforms and identify system requirements.

System Design (5–6 days) Design system architecture, database schema, and APIs.

Development (10–12 days) Implement lecture creation, user enrollment, lecture streaming access, and question submission.

Testing (4–5 days) Perform unit testing and integration testing.

Refinement (3–4 days) Fix issues, optimize performance, and prepare documentation.

Estimated time for prototype development: **approximately four weeks**.

4 Domain

This project belongs to the **Education Technology (EdTech)** domain. The platform improves digital learning environments by enabling distributed classrooms where students from multiple universities can attend lectures delivered by leading professors. By addressing scalability and interaction challenges in large online lectures, the system demonstrates how modern software architectures can support collaborative and accessible education at scale.

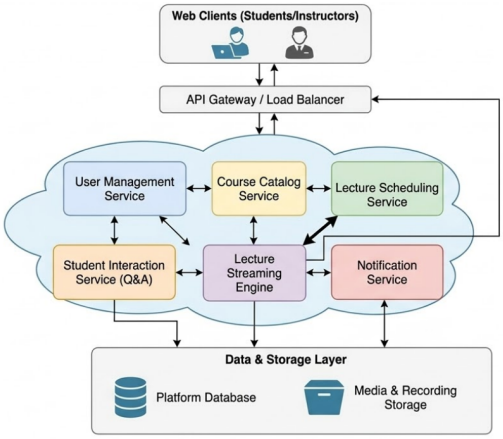

Figure 1: Proposed Microservices Architecture